ModelIt!

Teacher's Guide

Why Do Animals Live in Groups?

The Power of Teamwork in Nature

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-LS2-1

Construct an argument that some animals form groups that help members survive.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students construct an argument supported by model evidence that animals forming groups increases survival rates through improved predator detection.
Disciplinary Core Idea	**LS2.D Social Interactions and Group Behavior** — Being part of a group helps animals obtain food, defend themselves, and cope with changes. Groups may serve different functions and vary dramatically in size.
Crosscutting Concept	**Cause and Effect** — Students investigate how changing group size (cause) affects predator detection and survival rate (effects) in a predictable, positive relationship.

CAST Testing Alignment

MULTIPLE CHOICE:

A herd of 30 antelope spots a lion sooner than a single antelope standing alone. Which best explains why the herd detected the lion faster?

CONSTRUCTED RESPONSE:

A student says that animals should live alone because groups attract more predators. Using evidence from your model, explain why living in a group is actually SAFER for most animals. Use the words 'group size' and 'predator detection' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

From tiny ants to enormous elephants, millions of animal species choose to live in groups. This is not random — group living provides real survival advantages that scientists have studied for decades.

Why Do Animals Form Groups?

Animals form groups for several important reasons. The most common is protection from predators — more animals means more eyes and ears watching for danger. Groups also help animals find food more efficiently, stay warm by huddling together, care for young by sharing babysitting duties, and even hunt more successfully. Different species form different types of groups — herds, flocks, packs, schools, and colonies.

Safety in Numbers

The 'many eyes' hypothesis explains one of the biggest advantages of group living. In a group of 50 meerkats, while some eat, others stand guard as sentinels. If any sentinel spots a hawk or jackal, it gives an alarm call and everyone dashes for cover. A lone meerkat must both eat AND watch for predators — it cannot do both well. The dilution effect also helps: even if a predator attacks, each individual's chance of being caught is only 1 in 50 instead of 1 in 1.

Costs of Group Living

Group living is not all positive. Large groups need more food, which can lead to competition. Diseases spread faster in crowded groups. Some animals must share territory and mates. And large groups are easier for predators to FIND, even though individual members are harder to catch. Despite these costs, for most social animals, the survival benefits of group living far outweigh the drawbacks.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify group size as an external component and predator detection and survival rate as internal components that change in response to how many animals are in the group.

E **Establish Relationships**
Students determine that group size positively affects predator detection, and predator detection positively affects survival rate — more animals means faster warning means more survive.

V **Visualize & Model**
Students build a model showing how increasing group size leads to better predator detection and higher survival rates for the group.

E **Evaluate Outcomes**
Students run alone, small group, and large herd scenarios to observe how changing group size affects detection and survival across the system.

R **Revise & Extend**
Students add food availability or warning signal speed to explore tradeoffs of group living — bigger groups are safer but need more food.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Imagine you are a zebra on the African plains. Would you rather be alone or with a hundred friends? Why?"*

Watch For

Let students share their answers. Show images of animal groups: a wolf pack, a school of fish, a flock of birds.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to figure out why SO many animals choose to live in groups instead of going solo!"*

Watch For

Read objectives together. Introduce vocabulary with examples: a lion is a predator, a zebra uses group behavior for survival.

Slide 3: The Big Question

3 min

Say: *"Why do animals live in groups? Is it just because they like having friends, or is there a survival reason?"*

Watch For

Play a quick game: one student closes their eyes while others watch for a 'predator' (teacher holding a picture). Which spots it first?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to prove whether groups really do help animals survive!"*

Watch For

Preview each LEVER step. Explain that scientists use models to test ideas about animal behavior.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of this system can we identify? What can we change, and what happens as a result?"*

Watch For

Guide sorting of three components. Explain group size is external because the number of animals is what we are testing.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When there are MORE animals in the group, does predator detection go up or down? And what about survival?"*

Watch For

Help students draw positive arrows from group size to predator detection, and from detection to survival rate.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test it! First one animal alone, then a small group, then a big herd. What do you predict?"*

Watch For

Have students write predictions first, then run each scenario. Discuss: Were you surprised by any results?

Slide 8: What Did We Discover?

7 min

Say: *"The evidence is clear — groups save lives! Safety in numbers is REAL."*

Watch For

Summarize discoveries. Show a short video clip of meerkats taking turns as sentinels. Ask: How does this connect to our model?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Group Size
- **INTERNAL:** Predator Detection, Survival Rate

Relationships

- **Group Size to Predator Detection:** POSITIVE (+) — More animals in the group means more eyes and ears scanning for danger. A group of 50 animals is much more likely to spot a predator quickly than a single animal trying to watch in all directions alone.
- **Predator Detection to Survival Rate:** POSITIVE (+) — When predators are detected earlier, the group has more time to escape. Early warning gives animals a head start to run, hide, or defend themselves, which means more of them survive the encounter.

Key Discoveries

- Animals in bigger groups detect predators faster because more eyes are watching for danger
- Higher predator detection leads to higher survival rates — early warning saves lives
- Group living has other benefits too, like sharing body heat, finding food together, and protecting babies
- Even though groups attract attention, the safety in numbers advantage usually outweighs the risk

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals live in groups just because they are friendly"	Group living is primarily about survival, not friendship. Animals form groups because it increases their chances of detecting predators, finding food, and staying safe. While some animals do form social bonds, the main reason for grouping is survival advantage.
"Bigger groups are always better for every animal"	Very large groups have disadvantages — they need more food, attract more predators, and diseases spread faster. Each species has an ideal group size that balances safety with food competition. Lions live in small prides while wildebeest form herds of thousands — each size fits that animal's needs.
"Predators only attack animals that are alone"	Predators attack groups too, but they have a much harder time. Predators often target the edges of a group or try to separate one animal from the herd. The confusion of many animals running in different directions makes it harder for a predator to focus on one target.